Advanced Web Application Development Technologies

Nikolaos Korfiatis & Dimitris Ringas

nkorf@ionio.gr | riggas@ionio.gr

Department of Informatics

Ionian University



The course page

https://opencourses.ionio.gr/courses/DDI291/

there is also a github repo... (to be refreshed)

https://github.com/riggas-ionio/web-II/tree/class-2025



Contacting the instructors

- Instructors' email
  - Nikolaos Korfiatis - nkorf@ionio.gr
  - Dimitrios Ringas - riggas@ionio.gr

- Messages through the course page
- Discussions (opencourses tool)
- Emails that do not clearly state:
  - the course
  - your details (full name and student ID)
  - your question
  - will NOT be answered...

Lectures

- Monday 09:00 – 11:00 – ΑΙΘ3
  - Theoretical Lectures
  - Seminar Lectures
- Tuesday 11:00 - 13:00 - Galinos Lab
  - Lab Sessions
  - Project Support
- Wednesday 17:00 - 19:00 - TABM 2.45
  - Tutorial Sessions
  - Seminar Lectures

- The way groups will be split will be announced - depending on the number of participants

Course description



Course purpose

A comprehensive study of World Wide Web systems and modern web application development technologies, with emphasis on those focusing on the server side (backend), as well as topics related to security, data management and integration with the frontend.

Course objectives - Knowledge

- Recognize the principles of server-side web application development.
- Recognize and describe the fundamentals of the PHP / Python / Javascript languages.
- Describe the operation and importance of HTTP, session management and databases in web application development.
- Understand the concepts and importance of web security and identity management (OAuth2).
- Explain the SaaS architecture and the importance of Cloud Computing in application development.
- Distinguish the differences between SaaS applications and traditional web applications.
- Know the differences between client side and server side technologies.
- Describe the main steps in designing a complete web application.
- Evaluate different and alternative methodologies for developing complete web applications.

Course objectives - Skills

- Design dynamic web applications using PHP / Python / Javascript.
- Develop dynamic web applications using PHP/Python/Javascript and MySQL.
- Compare functionality across different development technologies (PHP, Python, Javascript).
- Combine knowledge of PHP, Python, Javascript, MySQL and Cloud to develop complete web applications capable of communicating with other World Wide Web systems.
- Distinguish the backend and frontend parts of an application, evaluating the role of each part.
- Implement authentication and authorization systems in web applications.
- Develop customized SaaS solutions for different user or business needs.
- Leverage cloud services for hosting and managing web applications.
- Apply security and performance optimization techniques in backend applications.
- Collaborate with fellow students, communicate in a team environment to organize and distribute the workload, and take individual initiatives and responsibilities regarding the execution of team projects.

Course objectives - Attitudes

- Perceive the importance and impact of innovative web applications on various aspects of society
- Recognize problems and design and implement complete web applications based on the social, scientific and ethical issues of the case study.
- Recognize the value of ensuring security, privacy, robustness and resilience of the system and applications in situations of unforeseen and malicious use.

Bibliography

- Book [122075105]: Fundamentals of Web Development (Greek edition), 3rd Edition, Randy Connolly, Ricardo Hoar
- Book [133028118]: Learning PHP, MySQL & JavaScript - A Step-by-Step Guide to Creating Dynamic Websites (Greek edition), Nixon R.
- Book [68387584]: PHP and MySQL Web Development (Greek edition), 5th ed., Thomson Laura, Welling Luke
- Book [102125023]: Technologies and Programming on the World Wide Web (in Greek) - 2nd Edition, Christos Douligeris, Roza Mavropodi, Evi Kopanaki, Apostolos Karalis
- Book [133024850]: Python for Everybody (Greek edition), SEVERANCE R. CHARLES
- Book [143562665]: Learn Python Easily (in Greek), Karolidis A. Dimitrios

- Kallipos: Web Programming (in Greek), Nikolaos Avouris, Christos Sintoris, ISBN: 978-618-228-104-8, 2023, Kallipos, Open Academic Editions

Course content



Course contents

- Introduction to server-side web application development.
- Basic principles of REST architectures and RESTful APIs. Introduction to server-side programming languages: PHP, Python, Node.js
- Fundamentals of the PHP language and Javascript.
- Using Python for web application development.
- HTTP requests and data handling. Database management. Advanced PHP features.
- The Model-View-Controller (MVC) architecture. Advanced Python features.
- Comparison between monolithic applications and microservices. Development of dynamic web applications.
- The HTTPS protocol and SSL/TLS certificates. Development of Authentication & Authorization systems. Session management techniques.
- Performance and optimization principles (caching). Connecting frontend with backend and API integration.
- DevOps and Automation in the Backend. Introduction to continuous integration and continuous deployment (CI/CD). Containerization technologies (e.g. Docker) and container orchestration (e.g. Kubernetes).
- Introduction to cloud computing and SaaS. Creating and deploying SaaS applications.
- The MQTT protocol. Message queue and event-driven architectures.
- Introduction to Search Engine Optimization. Advanced web application security topics.

Course contents - enriched (for this year)

- Client-side programming
  - HTML, HTML5
  - CSS
  - JavaScript
- Web crawlers
- Metadata on the World Wide Web: XML, JSON

Lab topics

- JavaScript
- Python
- PHP
- Apache Web Server / Flask
- MySQL / SQLite / InfluxDB
- JSON
- RESTful services
- Web APIs

Course assessment



Assessment

- Group Project
  - <=2 people
  - Presentation of the Group Project
  - Individual Oral Examination at a practical level
  - 50% of the grade
  - A grade >=5 is required to pass the course
- Final Written Exams
  - 50% of the grade
  - A grade >=5 is required to pass the course
- Plagiarism = 0 and referral to the General Assembly

Questions



Have a good and creative semester!

